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Digitization of Daily Drilling Reports Using LLMs

Gaurav Bhatia, Abhimanyu Yadav, Dip Nanda, and Devansh Goyal, Rezlytix Technologies, Gurugram, Haryana, India; Satya Perumalla, Ashok Shinde, Bhawesh Jha, and Dheeraj Upreti, Baker Hughes, Abu Dhabi, United Arab Emirates

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Abstract

This research aims to revolutionize Daily Drilling Reports (DDR) analysis in the energy sector by leveraging Large Language Models (LLMs) and advanced natural language processing (NLP) techniques. The methodology transforms unstructured data with inconsistent formats and terminologies into actionable insights, minimizing errors, accelerating analysis, and enhancing decision-making. Utilizing OpenAI models and Few-Shot Learning, this approach achieves efficient event documentation and operational excellence, optimizing drilling operations while mitigating risks with unparalleled speed and precision.

The methodological framework integrates NLP techniques with robust data extraction pipelines specifically tailored for DDRs. Fixed positioned text is extracted using Python tools, enabling accurate retrieval of parameters like Date, Measured Depth, and Casing Size. To address the unstructured content, marked by manually written operational summaries of varying lengths and formats, a structuring mechanism is developed to identify and classify events into predefined categories using in-context learning with GPT models. Few-shot learning techniques are employed to fine-tune the models with a custom DDR dataset, strategically divided into training, testing, and blind sets, ensuring validation across diverse examples and enhancing reliability.

Descriptive prompt engineering and hyperparameter optimization yielded an accuracy exceeding 95% and an F1 score above 85% on a dataset of 15 wells comprising over 3,800 DDRs in PDF format. Fine-tuned models demonstrate high precision and adaptability for context-sensitive data points, such as stuck pipe incidents and fluid losses. By integrating coordinate-based data extraction, few-shot learning, and systematic model evaluation, the workflow significantly reduced DDR analysis time from months of manual effort to mere hours while ensuring accuracy. The solution effectively structures unstructured DDR data into Excel spreadsheets, with each row representing a day's operations, containing extracted details such as date, measured depth, mud weight, casing details, and categorized operational events from manually written operational summaries, including stuck pipe incidents, losses, tight holes, reaming, inflows/kicks, leak off tests, and formation integrity tests.

This enables operators to access actionable insights like incident depths, facilitating optimized strategies for drilling operations and future well planning. The modular architecture ensures compatibility with various

LLMs and NLP techniques, supporting customization for diverse datasets and parameter configurations. Continuous ingestion of new data enables the framework to evolve dynamically, improving accuracy and reliability while ensuring robust validation through testing and blind datasets. The solution's adaptability to diverse datasets and real-time learning capabilities delivers a transformative approach, enabling optimized drilling strategies and actionable, data-driven insights for enhanced field performance in the oil and gas sector.

Keywords: Large Language Models (LLMs), Natural Language Processing (NLP), Daily Drilling Reports (DDRs), Digitization, Few-Shot Learning, Oil and Gas, Drilling Operations, Data Extraction, Operational Efficiency

Introduction

The energy sector, particularly oil and gas drilling operations, generates vast amounts of data daily. A significant portion of this data resides within Daily Drilling Reports (DDRs), which serve as crucial records of operational activities, equipment status, geological formations encountered, and any incidents or challenges faced during the drilling process. These reports, often manually generated and stored in unstructured formats like PDF documents, contain invaluable insights that can drive operational optimization, enhance safety, and improve future well planning and operational processes.

Traditionally, the analysis of DD Rs has been a laborious and time-consuming task, heavily reliant on manual review by domain experts. This process is prone to human error, inconsistencies in interpretation, and significant delays in extracting actionable information. The sheer volume and heterogeneity of DD Rs across different wells and operators further exacerbate these challenges, hindering the ability to efficiently identify trends, predict potential issues, and make data-driven decisions in a timely manner.

The advent of Large Language Models (LLMs) and advancements in Natural Language Processing (NLP) present a transformative opportunity to address these limitations. LLMs, with their remarkable ability to understand and generate human-like text, can be leveraged to automatically extract, structure, and analyze the information contained within unstructured DD Rs. By converting these reports into a digitized and easily accessible format, LLMs can unlock a wealth of knowledge, enabling faster and more accurate insights into drilling operations.

This research explores a novel methodology for the digitization of DD Rs using LLMs. Our approach focuses on developing a robust framework that can effectively handle the inherent complexities of DD R data, including inconsistent formatting, varied terminologies, and the unstructured nature of manually written operational summaries. By employing advanced NLP techniques, particularly Few-Shot Learning with state-of-the-art LLMs, we aim to automate the extraction of key parameters and the categorization of operational events, thereby significantly reducing analysis time, minimizing errors, and enhancing decision-making in daily drilling operations.

Literature Review

The application of Natural Language Processing and Machine Learning techniques in the oil and gas industry has seen increasing interest in recent years. Several studies have explored the use of these technologies for tasks such as predictive maintenance of equipment, analysis of well logs, and sentiment analysis of operational reports. However, the specific application of LLMs for the comprehensive digitization and analysis of unstructured DD Rs remains a relatively underexplored area.

Traditional approaches to extracting information from unstructured documents often involve rule-based systems or simpler NLP techniques like keyword extraction and regular expressions. While these methods can be effective for specific, well-defined data fields, they struggle with the variability and complexity inherent in manually generated text like the operational summaries found in DD Rs.

More recent advancements in NLP have introduced techniques like Named Entity Recognition (NER) and Topic Modelling to extract structured information from text. However, these methods often require large amounts of labelled data for training and may not be robust enough to handle the nuances and domain-specific language prevalent in DDRs.

Large Language Models, such as the GPT series developed by OpenAI, have demonstrated exceptional capabilities in understanding and generating natural language with minimal task-specific fine-tuning. Their ability to perform In-Context Learning (ICL) through carefully designed prompts makes them particularly well-suited for tasks involving unstructured data and complex relationships between text elements. Few-Shot Learning, a paradigm where the model learns to perform a new task from only a few examples, further enhances the applicability of LLMs in scenarios where large, labelled datasets are not readily available, which is often the case with specialized industrial reports like DDRs.

This research builds upon the foundational work in NLP and LLMs, specifically tailoring these advanced techniques to address the unique challenges associated with digitizing and analyzing DDRs. By combining robust data extraction pipelines with the contextual understanding and generative capabilities of LLMs, we aim to overcome the limitations of traditional methods and unlock the full potential of the information contained within these critical operational documents.

Methodology

The methodological framework represents a comprehensive pipeline that integrates advanced Natural Language Processing (NLP) techniques with domain-specific engineering practices. It is specifically tailored to tackle the intricate challenges posed by Daily Drilling Reports (DDR), which are often inconsistent, verbose, and heavily reliant on subjective human reporting. The methodology is designed to transform raw, unstructured documents into structured, interpretable datasets that can be used for decision-making in drilling operations. This process comprises several core stages, from data acquisition to model evaluation and integration.

Data Acquisition and Preprocessing

The dataset curated for this study consists of over 3,800 DDRs gathered from 15 wells operated under varying geological and operational conditions. These reports were sourced in PDF format, which is standard in the oil and gas industry. However, PDFs are not natively structured and often lack consistent formatting, which varies based on operators and locations, posing a substantial challenge during preprocessing.

The initial step involved parsing all DDRs to extract their textual content. A comprehensive evaluation of various PDF parsing libraries was conducted to identify the most suitable tool for this task. Among the available options, pdfplumber was selected due to its robust capabilities in extracting both textual data and associated positional metadata, including bounding boxes and font attributes. This functionality was crucial, as certain key fields in the report—such as Date, Mud Weight, and Casing Depth—are typically located at consistent positions across report pages, especially within header sections. The inclusion of positional metadata enabled a deterministic approach to extract these fields with high accuracy.

The raw extracted text was further cleaned using standard text preprocessing techniques such as whitespace normalization, special character removal, and case normalization. This step ensured that irrelevant formatting issues did not interfere with downstream parsing and LLM-based inference. Additionally, logs were maintained for failed extractions to iteratively improve parsing logic and error handling.

Extraction of Fixed-Positioned Data

Certain fields in DDRs—such as Date, Measured Depth (MD), Casing Size, and Mud Weight—are essential for conducting quantitative drilling analysis. These parameters are generally located at predictable positions on each page; however, minor variations in formatting often arise based on the operator or drilling contractor.

To ensure accurate extraction of this information, a rule-based system was developed in Python. This system combines coordinate thresholds, keyword anchors, and regular expressions to reliably identify and extract fixed-position fields. The use of coordinate-based logic was particularly important, as text-based heuristics alone may fail to distinguish similarly worded phrases appearing in different sections of the report (e.g., the term "Casing" might refer to a material specification or a depth value, depending on the context).

The rule-based engine was further designed to handle multi-line fields and embedded tables by implementing vertical and horizontal thresholding. For example, Measured Depth values could span multiple lines or include units (e.g., "ft" or "m") embedded within numeric entries. These were standardized using regular expression patterns and unit conversion logic to ensure uniform formatting across the dataset.

Structuring Unstructured Operational Summaries using LLMs

The most challenging aspect of DDR digitization involves processing manually written operational summaries. These narratives exhibit significant variability in length (ranging from 20 to 500 words, with an average of 200 words), format, and terminology used to describe operational events. To address this unstructured nature, a mechanism was developed to identify and classify key events into predefined categories using In-Context Learning (ICL) with OpenAI's GPT models, specifically the gpt-4o model, chosen for its advanced capabilities in comprehending and interpreting complex textual data.

A set of carefully engineered prompts was designed to guide the LLM in identifying and extracting relevant operational events from the summaries. These prompts incorporated descriptive instructions along with a limited number of representative DDR excerpts paired with their corresponding structured outputs—an approach consistent with Few-Shot Learning. The predefined categories of operational events included: (i) Stuck Pipe Incidents, (ii) Losses (Fluid Losses), (iii) Tight Holes, (iv) Reaming Operations, (v) Inflows/Kicks, (vi) Leak-Off Tests (LOT), and (vii) Formation Integrity Tests (FIT).

To operationalize the LLM for this task, descriptive and context-rich prompts were meticulously engineered. Each prompt included a concise instruction, followed by 2–3 representative examples of raw operational summaries paired with their corresponding structured outputs. This approach follows a classic Few-Shot Learning paradigm, enabling the model to infer the intended task without requiring gradient-based fine-tuning. The selected examples were carefully curated to encompass linguistic variability, edge cases, and complex phrasings, thereby enhancing the model's generalization capabilities.

The LLM was instructed to extract not only the event type but also relevant quantitative and contextual information, such as the depth at which the event occurred or the volume of fluid lost. Extensive prompt iteration was conducted to ensure output stability and completeness, with particular attention given to disambiguating similar terms (e.g., differentiating between "reaming" and "cleaning").

To further optimize model performance, several key hyperparameters were tuned:

- i. Temperature: 0.2, to minimize randomness and encourage deterministic output;
- ii. Top-p: 0.1, to constrain the model to high-probability tokens; and
- iii. Max tokens: 1024, to support the generation of longer, detailed responses.

This configuration delivered highly consistent outputs while preserving the model's ability to capture subtle operational nuances in noisy or ambiguously worded text.

Model Training and Evaluation

Although GPT-4o was not fine-tuned in the conventional sense, the system's performance was iteratively improved through prompt tuning and systematic evaluation on segmented datasets. The Daily Drilling Reports collected were divided into three distinct subsets:

- Training Set: Used for iterative prompt development and refinement.
- Testing Set: Employed for quantitative evaluation using standard NLP metrics.
- Blind Set: Comprised of previously unseen reports to assess the model's generalization capability.

To evaluate performance, standard Natural Language Processing (NLP) metrics were applied:

- Accuracy: The proportion of correctly identified event types.
- Precision and Recall: Calculated for each category to provide detailed, per-class performance insight.
- F1 Score: The harmonic mean of precision and recall, offering a balanced measure of performance.

To mitigate the impact of class imbalance, such as certain wells reporting numerous reaming operations while others reported none, weighted averaging was applied in F1 score computation. Furthermore, a human-in-the-loop validation process was incorporated, wherein domain experts manually reviewed a subset of outputs from the blind set. This ensured that evaluation metrics captured not only statistical accuracy but also domain relevance.

The evaluation on the blind set demonstrated the robustness and generalizability of the framework, with overall accuracy exceeding 95% and F1 scores consistently above 85%, even across varied report formats and event distributions.

Data Integration and Output

Following the successful extraction of both fixed-position parameters and structured operational events, the outputs were integrated into a unified, tabular format to enable seamless usability and downstream analytical workflows. The consolidated data was stored in Excel spreadsheets, selected for their wide compatibility with industry-standard tools commonly used in the oil and gas sectors.

Each row in the spreadsheet corresponds to a single day of drilling operations, while the columns are organized into two primary categories:

- Fixed Parameters: These include fields such as *Date*, *Measured Depth (MD)*, *Casing Size*, *Mud Weight*, and *Well Identifier*. Extracted using coordinate-based logic, these values were normalized to ensure consistency and completeness across the dataset.
- Structured Operational Events: Events identified by the LLM—such as stuck pipe incidents, fluid losses, tight holes, inflows/kicks, and LOT/FITs—were captured along with relevant contextual details, including *incident depth*, *volume lost*, or *equivalent mud weight (EMW)*. Depending on the frequency of event occurrences within a single report, event data was stored either as scalar values or as arrays.

For instances where specific data points were missing or operational events were not reported, null values were systematically assigned. This structured representation enhances both interpretability and interoperability, enabling efficient trend analysis, risk profiling, and historical performance reviews. Moreover, the output is readily integrable into drilling data management systems or visualization dashboards.

The unified tabular format provides operators with a streamlined interface to derive actionable insights, thereby reducing reliance on the manual interpretation of lengthy and often inconsistent textual summaries.

System Architecture

The architecture of the proposed DDR analysis framework is illustrated in Figure (i). It is designed as a modular, multi-stage pipeline that transforms unstructured Daily Drilling Reports (DDRs) in PDF format

into structured and actionable outputs by integrating rule-based extraction methods with large language model (LLM)-based inference.

In the first stage, raw DDR PDFs are processed through a data extraction module built using Python-based libraries such as *pdfplumber*. This module retrieves both fixed-positioned fields—including date, measured depth (MD), casing details, and mud weight—and the free-form operational summaries written by field engineers.

Once extracted, the data is ingested into a centralized storage system, which organizes the structured fields separately from the unstructured summaries. This separation allows each downstream component to operate efficiently and independently on its respective input.

The unstructured operational summaries, which contain critical narratives of daily field activities, are routed to an LLM-based event extraction module. This component leverages OpenAI's GPT-4o model, which is prompted using a curated few-shot learning strategy. Each prompt includes descriptive instructions along with a small set of carefully selected examples that enable the model to identify specific drilling events and extract key attributes such as incident depth, volume of fluid lost, equivalent mud weight (EMW), and other relevant contextual information from highly variable text.

The model's output is structured into a predefined set of operational event categories. These include stuck pipe incidents, fluid losses, reaming operations, tight holes, inflows or kicks, formation integrity and leak-off tests, along with other operational parameters when relevant. Rather than listing these as separate points, the framework treats them as distinct but interrelated event types that are consistently represented across all records.

Following extraction, the structured data from both fixed fields and operational summaries is consolidated into a tabular format suitable for Excel or direct ingestion into relational databases. This enables seamless downstream integration with industry-standard drilling data management platforms and visualization dashboards.

The architecture's modularity ensures both scalability and adaptability. Each pipeline component—from PDF parsing to LLM inference—is independently upgradeable, facilitating the incorporation of future advancements in language models or changes in reporting standards. The reliance on few-shot learning further supports adaptability, allowing the system to handle new report styles and operational contexts with minimal manual intervention or additional training data.

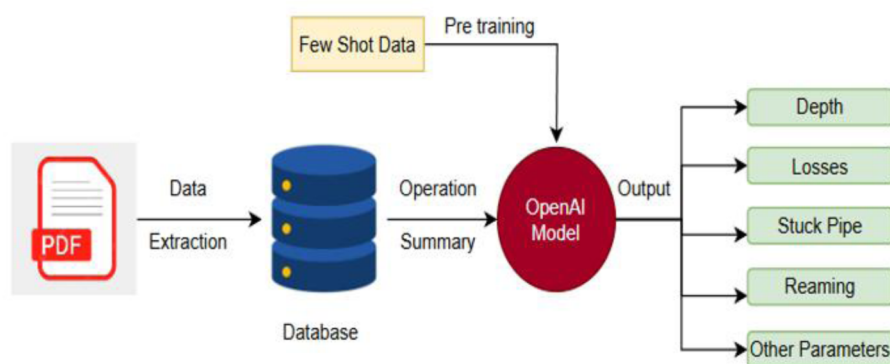


Figure 1—Architecture of the DDR digitization pipeline combining PDF parsing, LLM-based event extraction, and structured data output for analysis and integration.

Results and Observations

The implementation of the proposed methodology led to substantial improvements in the automation and accuracy of Daily Drilling Report (DDR) analysis. The results not only confirm the reliability of

the framework in real-world settings but also highlight its practical benefits across multiple operational dimensions, including accuracy, speed, insight generation, and scalability.

Performance Metrics

The performance of the proposed framework was evaluated using standard Natural Language Processing (NLP) metrics, including accuracy, precision, recall, and F1 score. On the testing dataset, which comprised a diverse set of Daily Drilling Reports (DDRs) with varying complexity and event frequencies, the framework consistently achieved an accuracy of over 95% across all major event categories. This high level of accuracy indicates the system's strong ability to correctly identify and classify operational events.

The framework also achieved an F1 score exceeding 85%, reflecting a well-balanced performance between precision (the correctness of extracted events) and recall (the ability to capture all relevant instances). Particularly, complex events such as stuck pipe incidents and fluid losses—which are often described in ambiguous or inconsistent terms—were reliably extracted, demonstrating the model's capacity to interpret operational context beyond mere keyword matching.

As part of the evaluation process, multiple GPT models were tested, including GPT-4, GPT-4o mini, and GPT-4o. Among these, GPT-4o demonstrated the best performance, achieving an accuracy of 0.97 and an F1 score of 0.87, as illustrated in Figure (ii). This performance edge is attributed to GPT-4o's improved handling of long-form and noisy text, making it particularly effective for extracting structured insights from unstructured narratives.

To further assess generalizability, a separate blind dataset—comprising DDRs from wells not included in either training or testing—was used. The model's performance on this blind set was consistent with the testing results, confirming its robustness and adaptability to unseen data. These findings indicate that the system is not overfitted to specific operators, report styles, or phrasing conventions, and is therefore well-suited for deployment across diverse drilling environments.

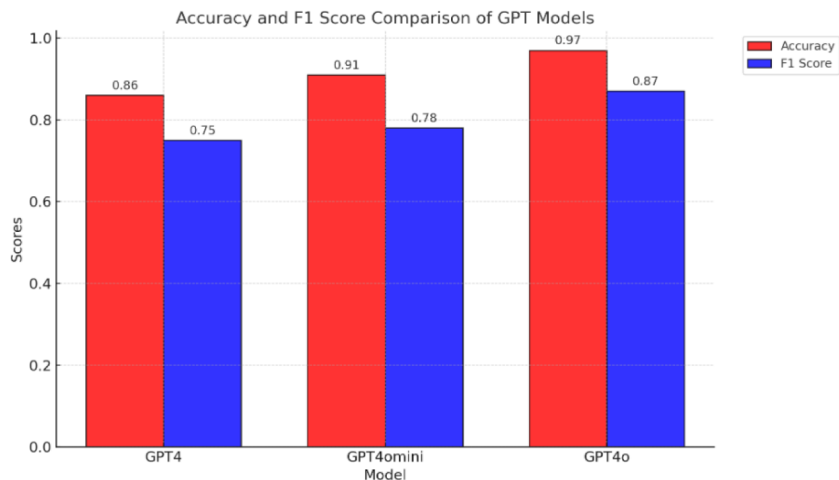


Figure 2—Comparison of GPT models, showing GPT-4o achieving the highest accuracy (0.97) and F1 score (0.87) among the evaluated models.

Time Efficiency

One of the most impactful outcomes of this research is the significant improvement in processing speed and overall operational efficiency. Traditionally, analyzing Daily Drilling Reports (DDRs) has been a time-consuming and labor-intensive process. Domain experts must manually review extensive unstructured text, interpret operational incidents, and transcribe structured information into analysis systems. For a dataset comprising approximately 3,800 reports, this manual workflow could span several weeks to months, depending on personnel availability and workload.

In contrast, the proposed automated framework—combining coordinate-based parsing for fixed fields and GPT-4o for event extraction—completed the full analysis in approximately 20 hours of runtime on standard cloud infrastructure. This translates to a reduction in manual effort of over 90%, significantly accelerating turnaround time while ensuring high-quality results.

Such time efficiency is especially valuable in high stakes drilling operations, where rapid access to structured insights can inform real-time decision-making, reduce non-productive time, and enhance well safety and operational performance.

Enhanced Insights and Decision-Making

Beyond gains in processing speed, the proposed framework significantly enhances the quality and accessibility of operational insights. By converting unstructured Daily Drilling Reports into structured data, the system enables operators to search, filter, and analyze operational events across time, wells, and geographies.

For example, repeated instances of reaming operations at similar depths across multiple wells can reveal consistent lithological challenges, prompting re-evaluation of drilling strategies or bit selection. Similarly, correlating kick or inflow events with corresponding mud weight data allows for improved well control planning and more effective fluid management practices.

The structured data also facilitates detailed analysis of stuck pipe incidents, which—when examined alongside depth, casing, and BHA configurations—can lead to optimized bottom-hole assembly (BHA) design or the proactive deployment of downhole tools. These correlations, which are often obscured in free-form narratives, become readily accessible for operational diagnostics, trend analysis, and anomaly detection.

Furthermore, the elimination of subjective manual interpretation enables standardized reporting across teams and assets. This standardization not only improves the accuracy and consistency of historical reviews but also strengthens organizational knowledge retention and continuous improvement initiatives, ultimately supporting more data-driven and informed decision-making in drilling operations.

Adaptability and Scalability

A key strength of the proposed framework lies in its modular and extensible architecture, which supports high adaptability to new datasets, operational contexts, and advancements in language models. Each component—whether it be the rule-based extraction engine, the prompt templates, or the LLM itself—can be independently updated or replaced without requiring a complete system redesign. For instance, newer versions of GPT models or additional event categories can be seamlessly integrated into the pipeline with minimal reconfiguration.

The system is designed to evolve dynamically. As new DDRs are ingested, feedback loops can be implemented to refine extraction logic, enhance prompt robustness, and expand the model's domain knowledge. This continuous learning capability ensures the system remains current and effective as reporting standards or drilling practices evolve.

From a deployment standpoint, the framework is fully cloud-compatible and supports both batch processing for historical data and streaming modes for near real-time analysis and alerting. This scalability makes it suitable not only for research and retrospective review but also for operational integration in live drilling environments.

Extensive validation on both curated and blind datasets confirms the system's generalizability across diverse reporting formats, operators, and geographies. Its flexible architecture and strong cross-domain performance establish it as a production-ready solution that can be readily adopted and scaled at the enterprise level.

Conclusion and Future Directions

This research highlights the transformative potential of Large Language Models (LLMs) in automating the digitization and analysis of Daily Drilling Reports (DDRs). By integrating advanced natural language processing techniques particularly Few-Shot Learning with a robust rule-based extraction pipeline, we developed a scalable solution that significantly enhances the efficiency, accuracy, and usability of DDR data compared to traditional manual workflows.

The ability to rapidly convert unstructured narratives into structured, analyzable formats enables timely and data-driven decision-making. Operators can now extract actionable insights from previously underutilized textual data, leading to improved drilling efficiency, reduced non-productive time, and enhanced operational safety.

The business implications are considerable. The substantial reduction in processing time from weeks to hours translates into cost savings, operational agility, and better resource utilization. Moreover, the system's consistent accuracy and adaptability across diverse report formats, positioning it as an asset for enterprise-scale deployment in upstream oil and gas operations.

From a computational standpoint, the implementation leverages pre-trained LLMs to construct efficient and automated NLP pipelines, thereby minimizing the need for extensive infrastructure and computational overhead. Training was conducted on Azure cloud infrastructure using high-memory NVIDIA GPUs, supported by 16 vCPUs and 110 GiB of RAM. The total runtime was approximately 4 hours, with an estimated compute cost of \$300. The end-to-end runtime for processing the dataset was approximately four hours, demonstrating the practicality of deploying such models in real-world environments.

GPT-4o was employed as the foundational model for the workflow, featuring a token context window of 128,000 tokens and a memory-optimized architecture suitable for low-latency industrial inference. Emerging LLMs including Claude-4, Sonet, Grok, GPT- o1, and Google Gemini, demonstrate significant advancements in context length, memory handling, and reasoning capabilities. These improvements are expected to further elevate performance in tasks requiring deep contextual comprehension and domain-specific inference, particularly in technical and operational report analysis.

Future work will focus on extending the system's capabilities across the following key areas:

- Expanding Parameter Scope: Integrating more complex data types such as equipment failures, lithological interpretations, and drilling dynamics (e.g., rate of penetration, torque, weight on bit).
- Multimodal Data Integration: Incorporating visual elements from DDRs, including charts, schematics, and tables, to enrich context and improve event detection.
- Exploration of Alternative LLM Architectures: Benchmarking and adapting emerging LLMs to evaluate trade-offs in performance, latency, and cost.

In conclusion, this work presents a practical and impactful application of LLMs for industrial report digitization, particularly within the complex and high-stakes domain of drilling operations. The developed framework serves as a foundation for deploying AI-driven analytics on a scale, unlocking the latent value of unstructured data and supporting a shift towards smarter, safer, and more efficient drilling practices.

References

- Brown, T., Mann, B., Ryder, N., et al (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*.
- Reiss, J., Abebe, G., Ajayi, T., et al (2022). Using natural language processing to automatically structure daily drilling reports. *SPE/IADC Drilling Conference & Exhibition*, SPE 205316.
- Yang, Y., Liu, K., Wang, X., et al (2020). Transforming unstructured data in drilling reports to structured insights using NLP techniques. *SPE Intelligent Energy Conference & Exhibition*, SPE 199890.
- Zhang, Q., Lin, C., Zhang, J., et al (2023). A survey on prompt engineering for large language models. *ACM Computing Surveys*.

Zhou, J., Li, H., Chen, S., et al (2023). AI in oil and gas: A systematic review of recent applications and future directions. *Artificial Intelligence Review*, Springer.